PRE – DISPATCH INSPECTION REPORT

Format No: QA-F-04/01(29-12-2020

Supplier:	M.S.INDUSTRIES	Customer :	MONO ENGINEERING LLC
Product Name :	BALL STUD	Vendor Code:	
Customer Part No. :	9636	Material Sp. :	EN1A L
Supplier Part No. :	MSI - TC-0011	Sample Size :	50 NOS

Invoice No:	MSI/2026.27
Invoice Date:	14/04/202
Lot Qty:	200 nos

SN	Parameters	Specifications	Tolerance	Class	Checking Method	MANUFACTURER'S OBSERVATIONS 1	2	3	4	5	Remarks	Cus Obse
1	TOTAL LENGTH	24.00MM	± 0.20 MM	---	VERNIER CALIPER	23.96	23.96	24.00	24.00	23.96	OK	
2	THREDING LENGTH	7.50MM	± 0.20 MM	---	VERNIER CALIPER	7.50	7.65	7.60	7.55	7.60	OK	
3	NUT HEIGHT	3.00 MM	± 0.30 MM	---	VERNIER CALIPER	3.00	3.05	3.08	3.00	3.05	OK	
4	THREADING	M8X1.25	---	---	M8X1.25 THREAD RING GAUGE	OK	OK	OK	OK	OK	OK	
5	WIDTH A/F	13.00MM HEX	- 0.27MM	---	VERNIER CALIPER	12.96	12.97	12.96	12.96	12.96	OK	
6	BALL DIAMETER	10 MM	- 0.20MM	---	VERNIER CALIPER	9.98	9.97	9.96	9.96	9.94	OK	
7	DIMENSION	7.20 MM	± 0.20 MM	---	VERNIER CALIPER	7.20	7.19	7.24	7.22	7.20	OK	
8	DIMENSION	10.70MM	± 0.20 MM	---	VERNIER CALIPER	10.70	10.71	10.72	10.70	10.72	OK	
9	SURFACE TREATMENT	ZINC CLEAR TRIVALENT PASSIVATION	8 - 10 MICRON	---	THICKNESS TESTER	8 µ	10 µ	8 µ	10 µ	8 µ	OK	
10	Surface:Free from Damage, Dent, Crack, Rusty, Scratch Marks.Burr & Sharp edges , Hole Shift,Plating not OK etc.				VISUAL	OK	OK	OK	OK	OK	OK	

SUPP. REMARKS: Lot Accepted ☑ / Rejected ☐ / Conditionally Accepted ☐

Date:

Inspection Date: 13/04/2026 Inspected By: PSP Approved By: M.S.INDUSTRIES

Customer Sign

Class : Important-I / Critical-C / Significant-S Parameter

Bhushan POWER & STEEL

BPSL -Depot Sambalpur WORKS

QR / QA / 54

(A JSW Group Company)

Works: Bhushan Power and Steel Limite,Sambalpur - 768200,Sambalpur, INDIA

Registered Office: "4th Floor, A-2", National Trust for Handicapped, USO, New Delhi, NEW DELHI - 110067

TEST CERTIFICATE

Certificate No. : 226412
Customer Name: Alloy Steel Mart
Cust. Address : B - 1002 LAKE FLORENCE
MUMBAI,Maharashtra-400076,India

Date : 06.09.2025
SO No : 402883636/80
SO Date : 26.08.2025
Truck No.: OD05BH0593

Po Number:Email
Po date :26.08.2025
Packing Slip No :719381320
Invoice No :0094121393
GST Invoice No : 25OD2112101417

Heat No	Grade	Size (mm)	Length (mm)	M\L (mm)	No. of Bundles	No. of PCS	Weight	Billet Size
A254533	EN1A(PB)	21.00 mm DIA			3		8.659 MT	180 X 180 mm

TDC NO : STD
Process Route : BF-EAF-LRF-VD-CCM (AMLC & EMS)-RM
Specification:

BIS Equivalent Grade :
Reduction Ratio : 93.51 : 1

CHEMICAL ANALYSIS

Prod STD No : EN1A(PB)

Element	C%	Si%	Mn%	P%	S%	Cr%	Mo%	Ni%	B %	V%	Cu%	Pb%	Sn%	Ti%	Nb%	Zr%	As%	Sb%	Al%	Ca %	CE%
MIN	0.0700		0.800		0.200							0.150									
MAX	0.1500	0.100	1.200	0.070	0.3000							0.350									
Actual	0.089	0.035	1.150	0.052	0.290	0.017	0.003	0.005	0.0000	0.0[illegible]	0.005	0.230	0.000	0.0000	0.000	0.000	0.000	0.000	0.002	0.0000	0.286

Gas Levels : N ppm : [/] -76 O ppm : [/] -63.0 H ppm : [/] -

MECHANICAL PROPERTIES :

	Tensile Test ( )				Hardness Test				Impact Test ( )			
Cond.					As rolled				Spec.			
Spec.	YS	UTS	Elong.	RA	As Supply	As Quenched		As Q&T	Test Temp	Imp. Value (min)	Avg.	
			%	%	BHN				°C	J	°C	J
Min												
Max												
Actual					118							

Upset Type : Upset Test : % Upset Result :

JOMINY HARDENABILITY ( )

Ideal Diameter (DI) : [/] -

Distance																				
MIN (HRC)																				
MAX (HRC)																				
Actual(HRC)																				

METALLURGICAL PROPERTIES & PHYSICAL INSPECTION

	%C.SEG	Step Down Test	Non-metallic Inclusion K3	MPI Test	Pearlite Amount	Spark Test	Non-metallic Inclusion K4	Anti Mixup Check(100%)
Actual						100% OK		
	Macro Etch Test	Micro Structure	CN	CZ Looser	CZ Closer	GBC Level	DEG. SPH	MFLT/ECT
Spec			/	/	/			
Actual	C1R1S1	FERRITE & PEARLITE						

	Grain Size	Decarb (mm)	Banding (µm)	A		B		C		D		DOS µm	Ultrasonic Test
				Thin	Thick	Thin	Thick	Thin	Thick	Thin	Thick		
Spec	/	/	/										
Actual	7.00				0.00	1.00	0.00	0.50	0.00	1.50	0.00		

Inclusion Rating Test Standard: ASTM_E45/IS_4163
Macro Etch Test Standard: ASTM_E381
Grain Size Standard: ASTM_E112/IS_4748
Banding Test Standard:
Decarb Standard: SAE_J419
UT Standard:

JSW COLOR CODE : BLACK+WHITE_DOT
CUSTOMER COLOR CODE : BLACK+WHITE_DOT

INSPECTION CERTIFICATE 3.1 ACCORDING TO EN 10204 : 2004

Supply Condition : AS ROLLED

13 mm Hex

Remarks :

Reference : WE HERE BY CERTIFY THAT THE MATERIAL SHIPPED UNDER THIS TEST CERTIFICATE DID NOT COME IN DIRECT CONTACT WITH ANY MERCURY, CADMIUM HEXAVALENT CHROMIUM CONTAINING DEVICES, FREE FROM RADIOACTIVE CONTAMINATION EMPLOYING A SINGLE BOUNDARY OF CONTAINMENT DURING THE MANUFACTURING PROCESS,TESTS, INSPECTION AND STORAGE.

WE CERTIFY THAT CONTENTS OF THIS REPORT ARE CORRECT AND ACCURATE AND MEET THE REQUIREMENTS OF THE PURCHASE ORDER,TECHNICAL DELIVERY CONDITION, GENERAL STEEL REQUIREMENTS ANDTHE MATERIAL CERTIFICATION REQUIREMENTS.

NOTE:
1. THE RESULTS RELATES ONLY TO THE ITEM TESTED.
2. CERTIFICATE SHALL NOT BE REPRODUCED EXCEPT IN FULL WITHOUT THE WRITTEN APPROVAL OF ISSUING AUTHORITY.

For BHUSHAN POWER AND STEEL LIMITED

ANIL KUMAR GOUDA

Date: 23rd February, 2026

Name Of The Customer : **M/s. M. S. Industries**
Alibaug.

ROHS Declaration

Dear Valued Customer,

This declaration is in regard to your request for information's on the Grauer & Weil (I) Ltd. Water based product lines listed below:

Product (s),

1.	GROWCHROME HB 157

The above mentioned products do not contain any hazardous ingredients, given below ~ Lead, Cadmium, Mercury, Hexavalent Chromium (CrVI), Polybrominated Biphenyl (PBB) & Polybrominated Diphenyl Ether (PBDE), Bis (2-ethylhexyl) phthalate (DEHP), Butyl benzyl phthalate (BBP), Dibutyl phthalate (DBPp), Diisobutyl phthalate (DIBP) stipulated in (EU) 2015/863 published on July 22, 2019 RoHS 3 directive 2015/863/EU, Annex II – as a part of the formulations of the above given our supplied products. The permissible limit of these specified 'hazardous' ingredients as per the said directive are also given below for ready refrence.

Sr. No.	Hazardous ingredients as per RoHS 3 directive 2015/863/EU.	Limits set by RoHS 3 directive 2015/863/EU. Annex II
1	Cadmium	100 ppm
2	Lead	1000 ppm
3	Mercury	1000 ppm
4	Hexavalent chromium (CrVI)	1000 ppm
5	Polybrominated Biphenyl (PBB)	1000 ppm
6	Polybrominated Diphenyl Ether (PBDE)	1000 ppm
7	Bis (2-ethylhexyl) phthalate (DEHP)	1000 ppm
8	Butyl benzyl phthalate (BBP)	1000 ppm
9	Dibutyl phthalate (DBP)	1000 ppm
10	Diisobutyl phthalate (DIBP)	1000 ppm

This declaration is issued to the best of our knowledge for the above mentioned product/s at the time of releasing this documentation.

For GRAUER & WEIL (INDIA) LIMITED

S. Chakraborty
SR. VP – R&D

Ashok Nalawade
GM – R&D

This information is believed to be accurate and refers to the laws, regulations and products at the date of issue. However, Grauer & Weil (I) Ltd. makes no express or implied representations or warranties with respect to the information contained herein. It is the responsibility of our customers to determine that their use of Grauer & Weil (I) Ltd. products is safe, lawful, and technically suitable for their applications. Because of possible changes in the laws and regulations, we cannot guarantee that the status of this product will remain unchanged.This declaration pertains solely to the specific finished products. Since it's a computerized printout, no signature is necessary for validity.

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www.growel.com
CIN L74999MH1957PLC010975

ALKHNANDA ENGINEERS.

AT- MHATROLI, POST- SARAL, ALIBAUG,RAIGAD-402201

PLATING TEST CERTIFICATE AS PER-ASTM-B117

TC. NO	AE: 036	CHALLAN NO	PARTY CH.NO	RECEIVED DATE
DATED:	06/04/2026	AE-86	45	04/04/26

CLIENT	M.S.INDUSTRIES PLOT NO. 9 & 10, SAHAJEEVAN VIVID UDYOG , SAGHTANA, ALIBAUG RAIGAD	MATERIAL DESCRIPTION: GREASE NIPPLE

SR.NO.	QUANTITY	PART DESCRIPTION	REQIREMENT	TEST RESULTS	PLATING	OBSERVED
1	5kg	Ball Stud	min 10 to 12 µ	Test observed 10µ	clear trivalent	ok
2	30kg	MSi-TA				
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						

VISUAL	free from inclusion ,patches blisters,chemicaloozing uncover and plattingdamage

INSPECTION BY:

FOR ALKHNANDA ENGINEERS

Jitu Naik

M/S. ALKHNANDA ENGINEERS

M.S. INDUSTRIES

PROPRIETOR

T.C check verified